Using Statcast Jump Metrics to Evaluate Outfield Defense

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What skills are most important to an outfielder’s ability to catch fly balls?

The Data

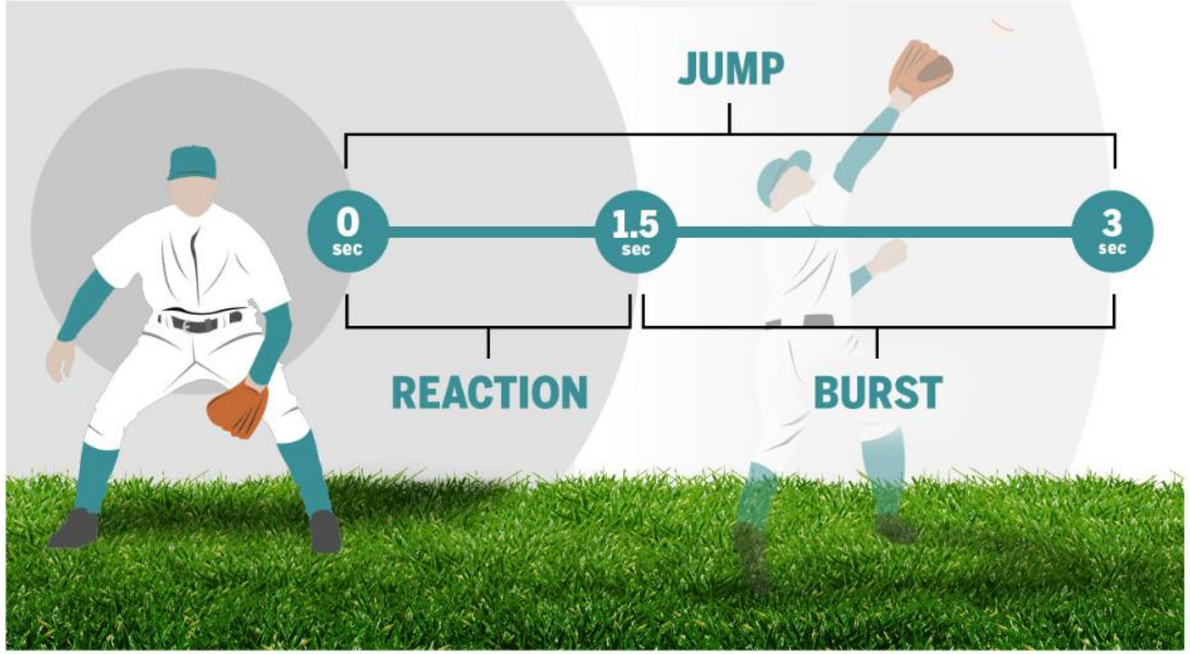
From Baseball Savant

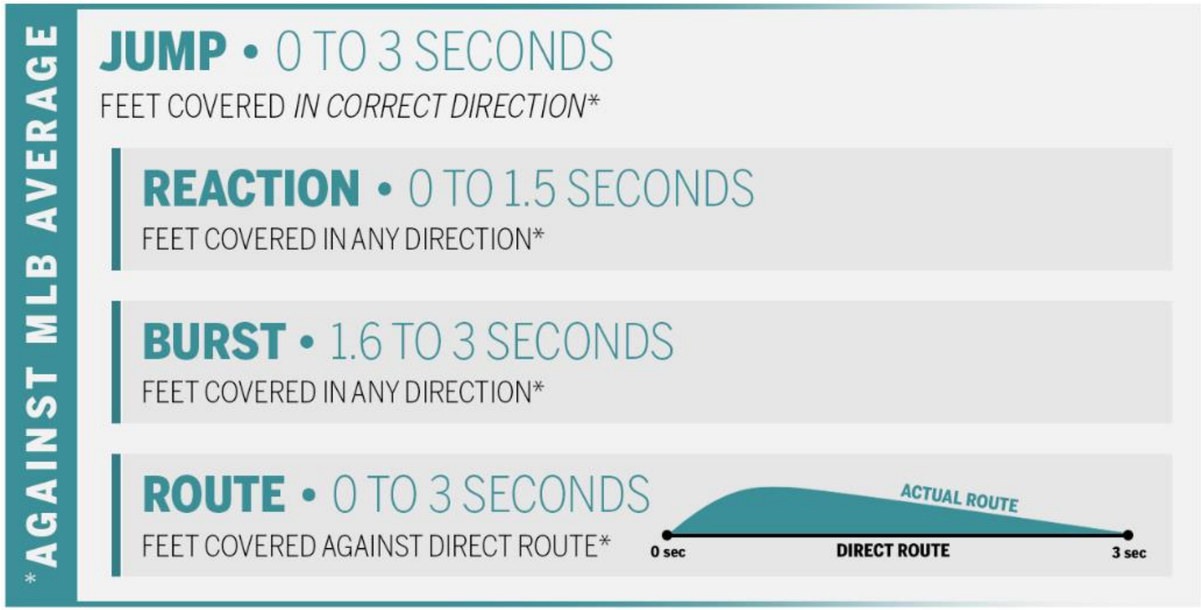
- Measured in ft relative to MLB average:
  - Reaction
  - Burst
  - Route
- Sprint Speed
- Play Difficulty (Stars)

From the Statcast API

- Distance
- Hangtime
- Catch/No Catch
- Play Difficulty (Catch Prob)

- *All data is from 2024*

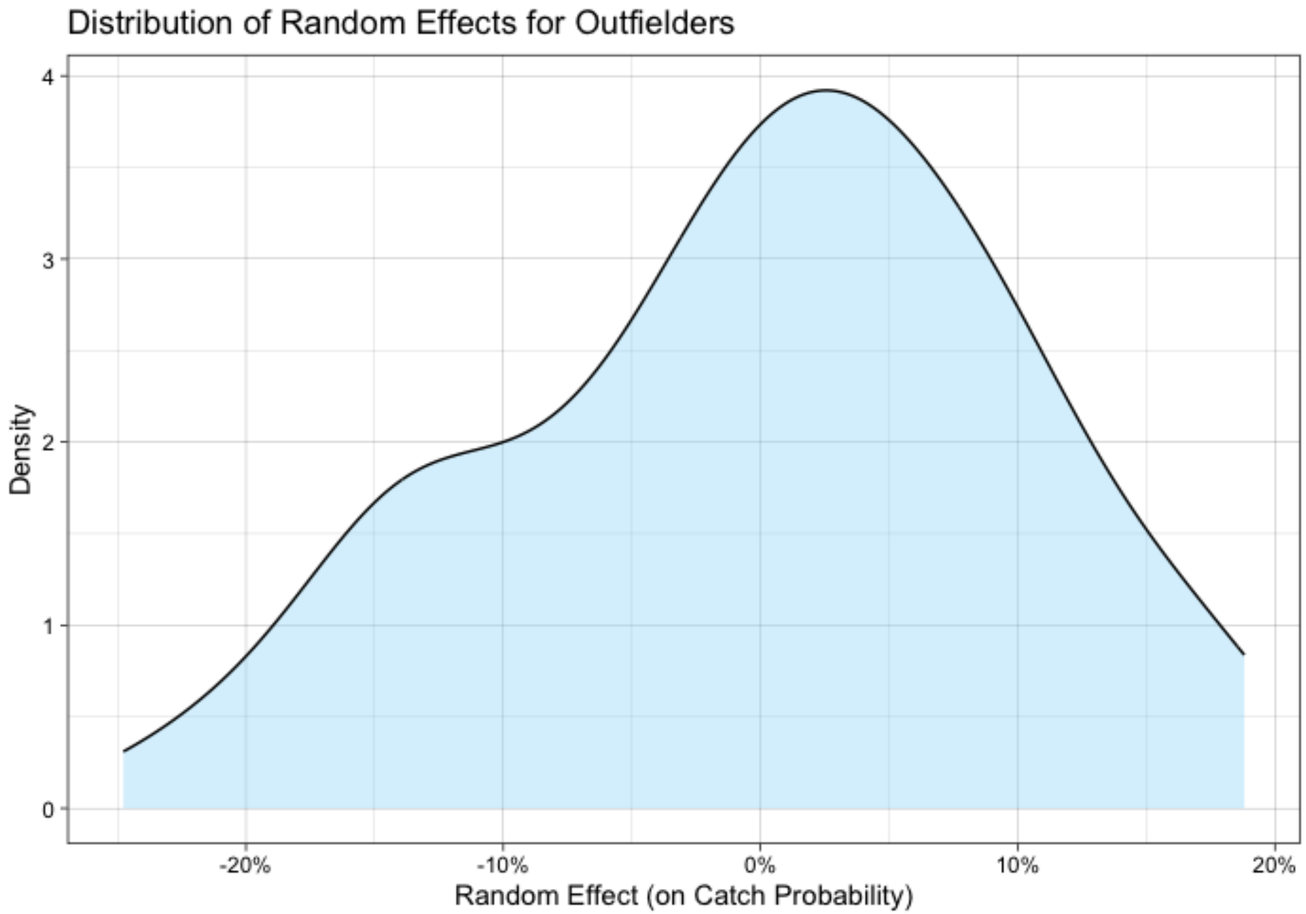




Fitting a Model

Logistic LMER (Linear Mixed Effects Regression)

- Modeling catch probability for each batted ball
- Fixed Effects
  - Mental Elements
    - Route
    - Reaction
  - Raw Athleticism
    - Sprint Speed
    - Burst
  - Difficulty of the specific batted ball
    - Distance to cover and hang time
- Random Effect: Represents player-specific variance not explained by other inputs

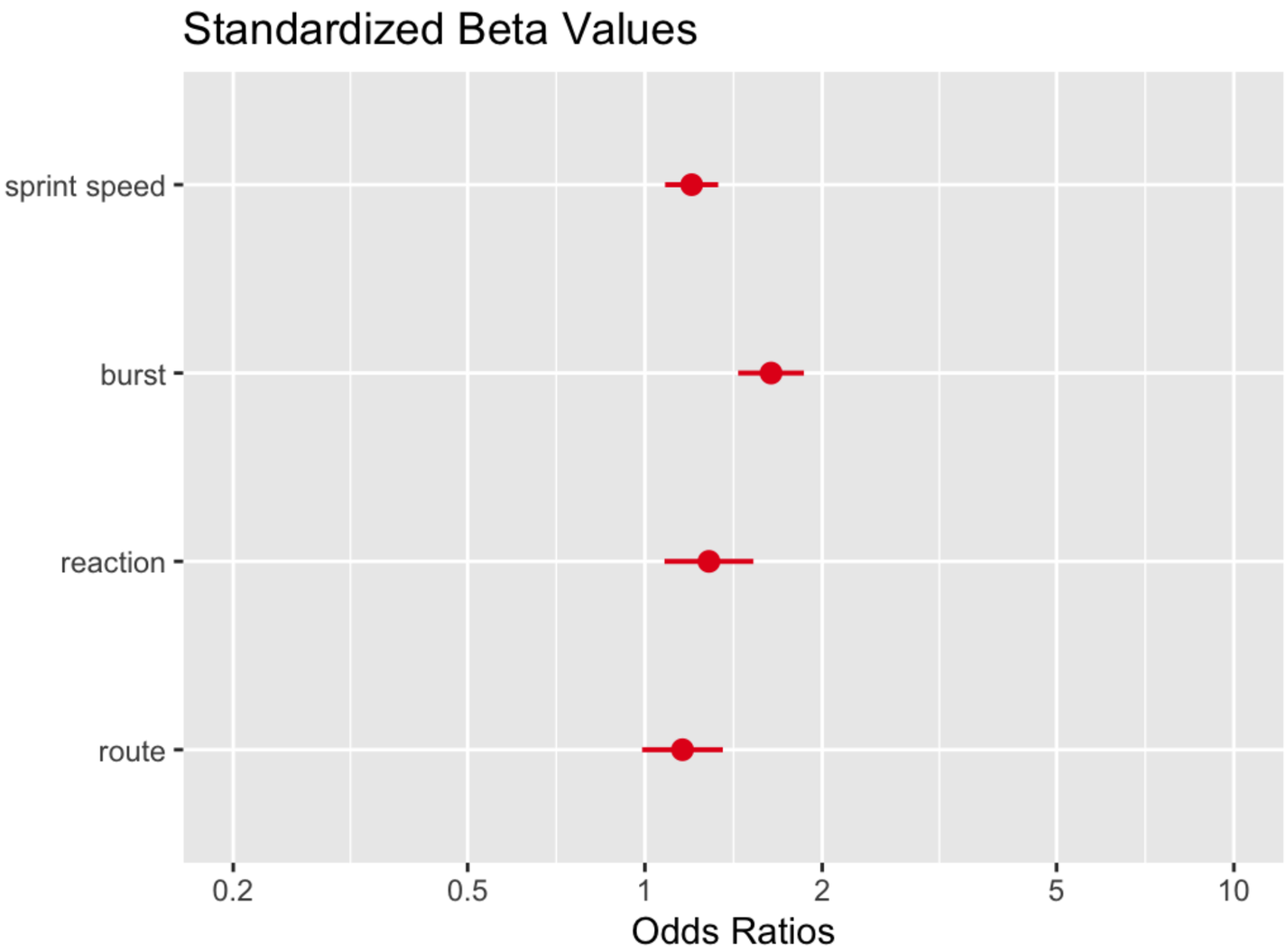


Lowest & Highest Random Effects

Player	Random Effect
Juan Soto	-24.7%
Wyatt Langford	-21.2%
Starling Marte	-20.1%
Cody Bellinger	-20.0%
Bryan De La Cruz	-16.8%

Player	Random Effect
Lars Nootbaar	18.8%
Wilyer Abreu	18.6%
Aaron Judge	18.3%
Anthony Santander	16.4%
Harrison Bader	16.3%

Fixed Effects



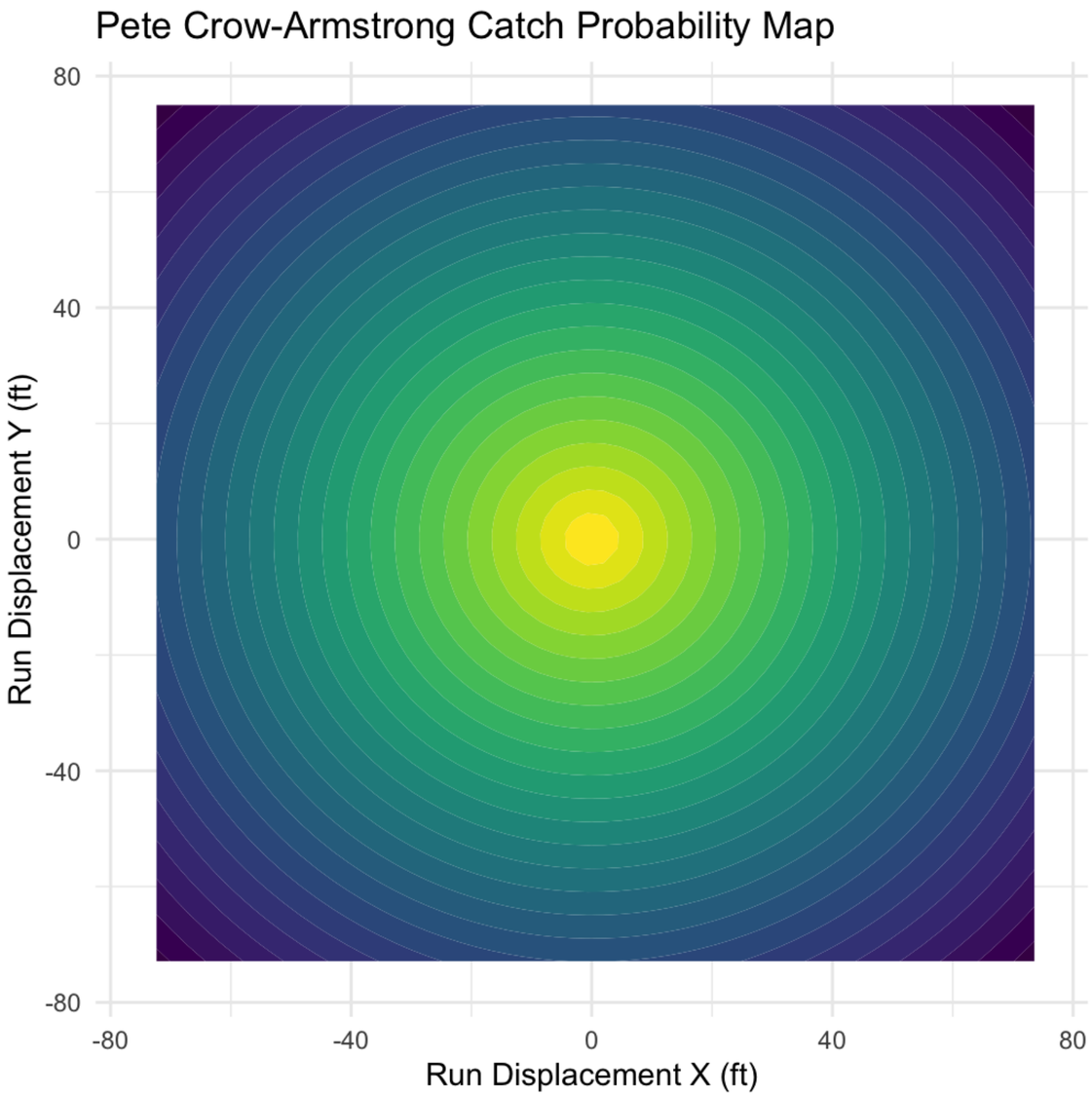
Applying the Model

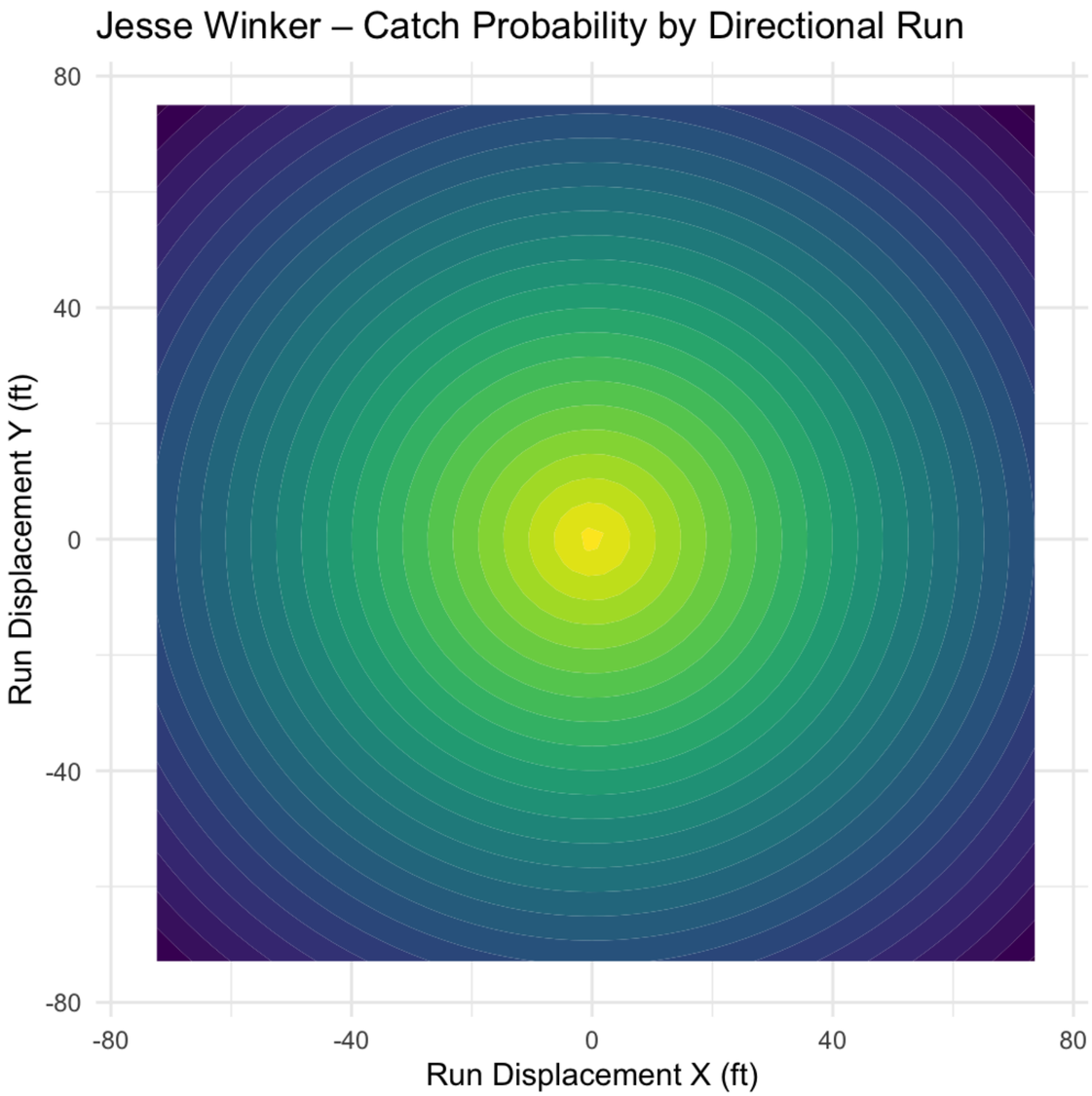
- Predicted whether each player would catch every batted ball in the dataset
- 5205 qualifying plays (Statcast catch probability under 90%)

Player	Percentage of Balls "Caught"
Jacob Young	70.9%
Jose Siri	70.5%
Kevin Kiermaier	70.2%
Pete Crow-Armstrong	69.6%
Michael Siani	69.4%

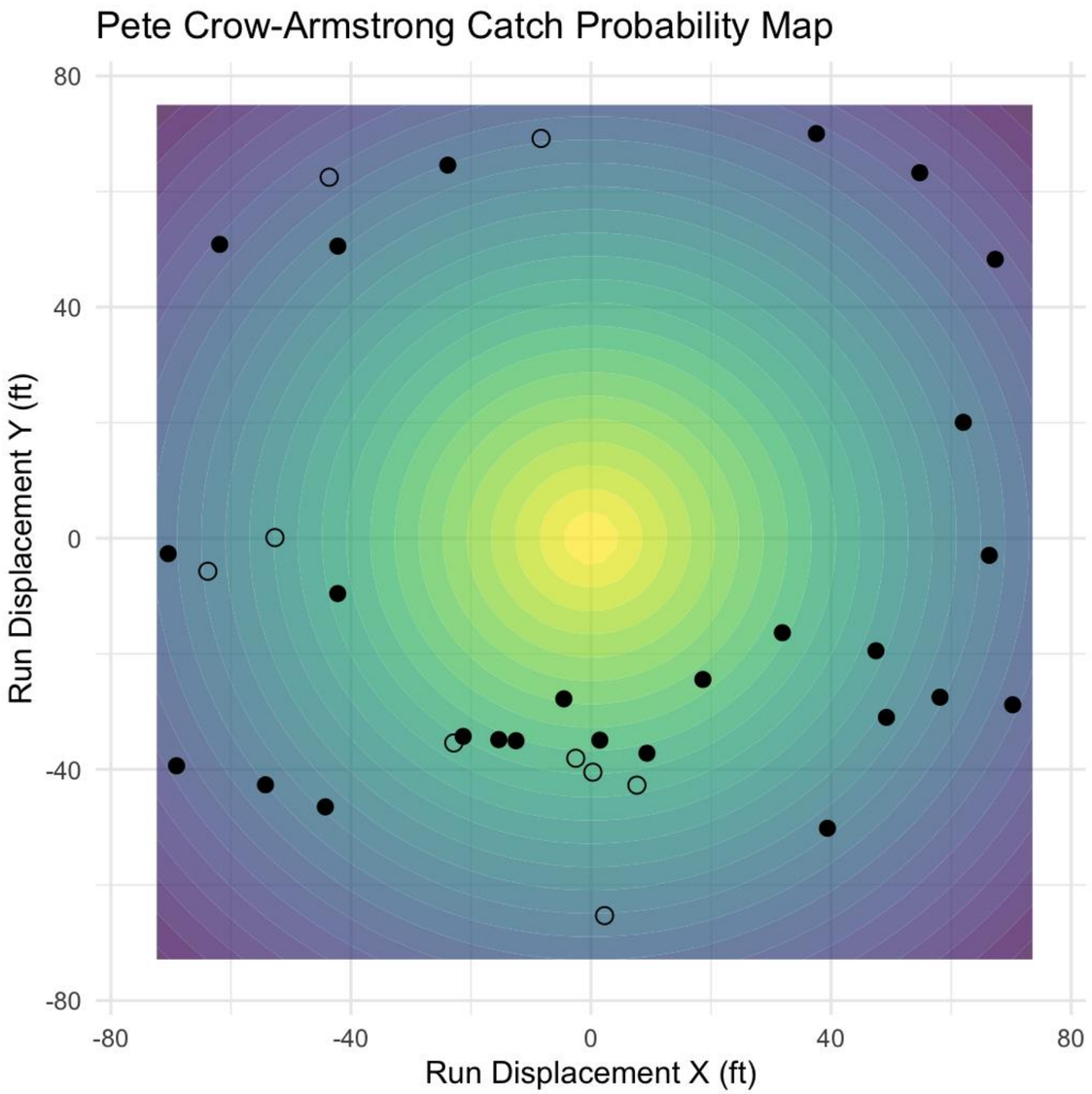
Player	Percentage of Balls "Caught"
Jessie Winker	18.1%
Nick Castellanos	20.2%
Aaron Judge	21.2%
Bryan De La Cruz	26.0%
Randy Arozarena	27.8%

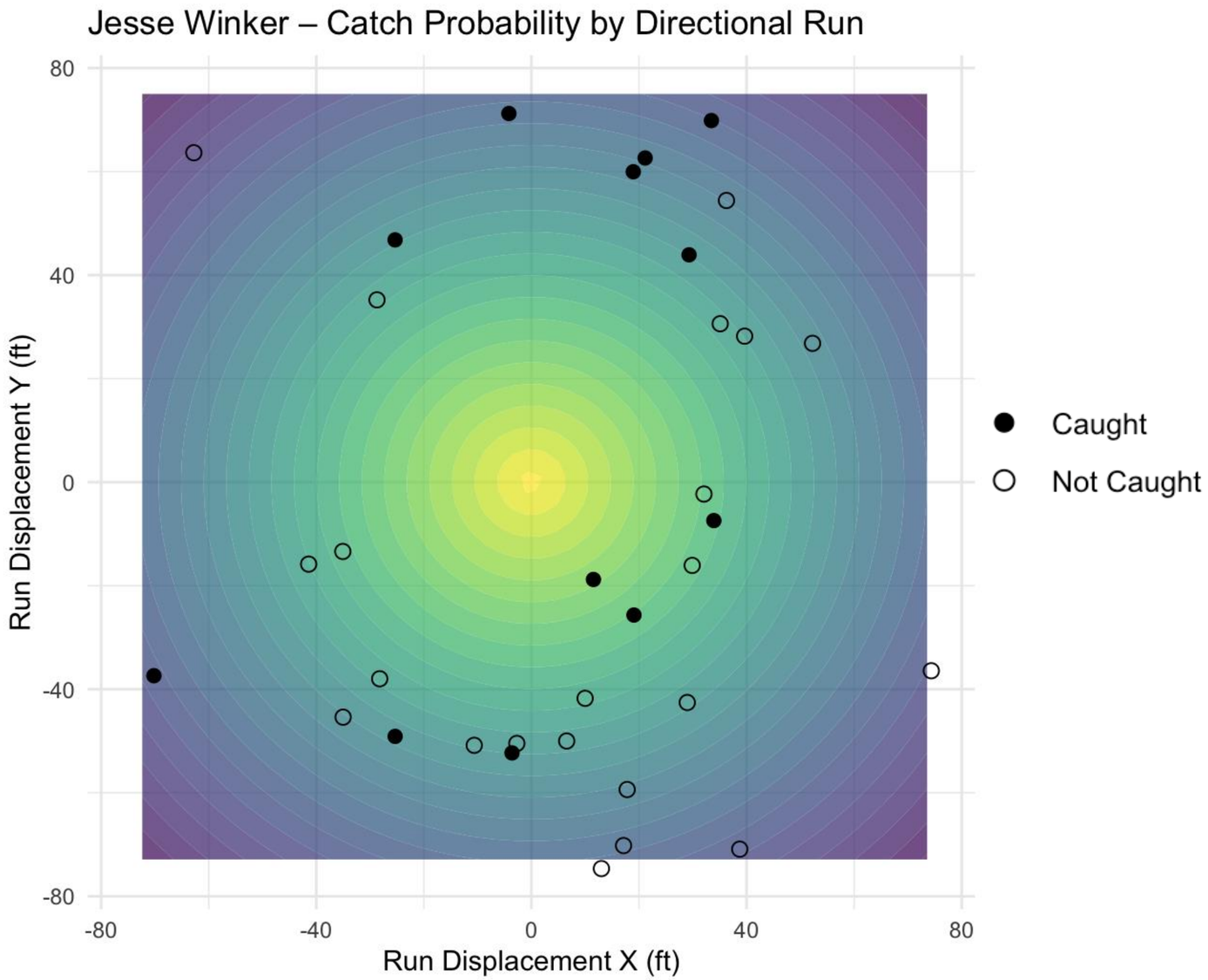
Mapping the Large Disparity





Contextualizing the Map/Disparity





Conclusions

- Burst (acceleration) is relatively more important than other aspects of jump or raw sprint speed
- Routes do not necessarily need to direct
- Players do not necessarily need to have elite speed

Thank you!

Questions?

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